Design Systems: Technical Worksheet

Grade 12 · Worksheet · About 25 minutes

Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

> **How to use this worksheet**
>
> - Do the parts **in order** from top to bottom.
> - Each part has a **time** so you know how long to spend.
> - Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: A **design system** is a shared library of **design tokens** (colors, spacing, type), reusable **components** (buttons, forms), **patterns**, and **guidelines**. It keeps products consistent and speeds up teams.

Do this:

1. **Name** two products by the same company that look similar and say what they share.

   _____

2. **Circle** the building blocks of a design system: Tokens and components · Random pictures

Part 2 – Tokens and components (about 5 minutes)

Design tokens store decisions as names and values (for example, color-primary = #0b5cad). **Components** use tokens so one change updates everywhere.

> **Real-life scenario:** The team changes color-primary once and every button in the app updates.

Discuss with a partner or reflect in writing: What could go wrong if every team made its own button style?

Do this: Circle the right layer.

Item	Layer
spacing-md = 16px	Token · Component · Pattern
Primary Button	Token · Component · Pattern
Checkout form flow	Token · Component · Pattern

Part 3 – Governance and documentation (about 5 minutes)

A system needs **documentation**, **versioning**, and **contribution rules**. Teams must know how to use, request, and update components.

> **Real-life scenario:** A new Date Picker is proposed, reviewed for accessibility, tested, documented, and released as v2.3.

Critical thinking: Why version a design system like software?

Word bank: *Documentation* · *Versioning* · *Review* · *Changelog*

Do this: Match each practice with its purpose.

Practice	Purpose
Explains how to use a component	_____
Tracks what changed	_____
Checks quality before release	_____

Part 4 – Scaling and trade-offs (about 5 minutes)

A rigid system speeds work but may limit creativity. A flexible one may drift into inconsistency. Teams balance **consistency**, **flexibility**, and **maintenance cost**.

> **Real-life scenario:** A team needs a special chart not in the library.

Discuss with a partner or reflect in writing: Should they build it locally or add it to the system? Justify.

Do this:

1. **Write** two criteria for adding a new component to a system.

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2. **Circle** the stronger approach: Contribute reusable parts back · Copy and tweak in secret

Part 5 – System map (about 5 minutes)

Do this: Draw a diagram with tokens → components → patterns → pages.

Write where a change in a token would show up.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can define **design tokens**.

☐ I can explain **components** and **patterns**.

☐ I know why **governance** matters.

☐ I can weigh **consistency vs. flexibility**.

☐ I can plan how to propose a component.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Design Systems (Technical). Central topic: Design systems at scale. Main technology area: UI design and front-end engineering.